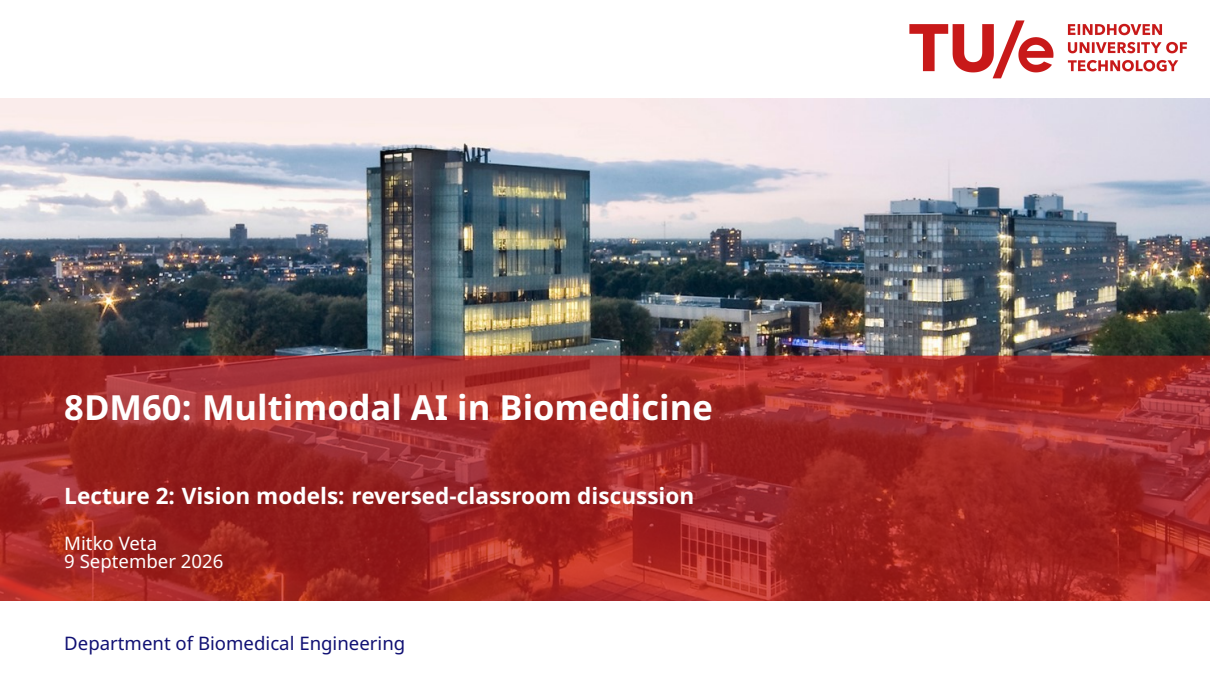




8DM60: Multimodal AI in Biomedicine

Lecture 2: Vision models: reversed-classroom discussion

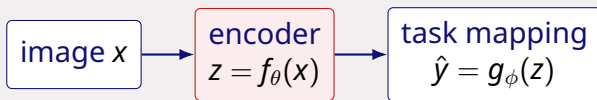
Mitko Veta
9 September 2026

Learning goals

By the end of this discussion, you should be able to:

- Describe key differences between task-specific vision models, visual foundation models and vision-language models.
- Distinguish image-only, contrastive image-text and generative image-text models by objective and output.
- Compare each general method with its biomedical counterpart.
- Critique evaluation, biomedical validation and limitations.

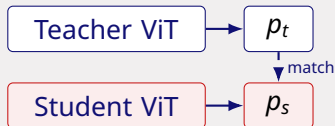
What is transferred?



Discuss: What changes when the encoder is pretrained for reuse rather than trained for one labelled task? Which part would you reuse for classification, segmentation or retrieval?

Distillation with No Labels (DINO, v2) in a gist

Two views of the same image



$$\theta_t \leftarrow m\theta_t + (1 - m)\theta_s$$

Moving average; m : momentum.

$$\mathcal{L}_{\text{match}} = - \sum_c p_{t,c} \log p_{s,c} \quad c: \text{learned prototype}; p_t, p_s: \text{output distributions.}$$

Student: gradient descent. Teacher targets: **stop gradient**.

Teacher centering/sharpening and KoLeo regularization counter collapse.

Source: [DINOv2, Section 4; teaching schematic, projection heads omitted](#).

Image level (DINO)

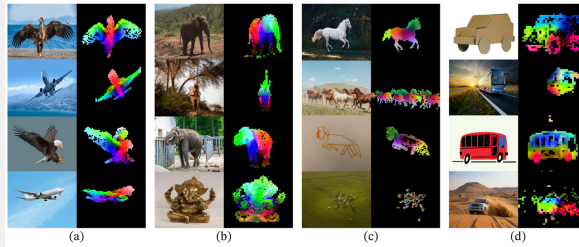
Match whole-image distributions across different crops of one image.

Patch level (iBOT)

Mask student patches; predict teacher targets at the corresponding visible locations.

Masked targets are features, not RGB pixels.

DINOv2: what do these features show?



No segmentation labels were used in pretraining. Colours show patch-feature PCA, not predicted class masks.

Discuss: How does this relate to Lecture 1's capabilities beyond the training objective? What would establish useful segmentation performance?

Source: [Oquab et al., Figure 1; figure only.](#)

DINOv2 and UNI: what changes in pathology?

General visual pretraining

Diverse images; reusable image and patch representations.

Discuss: What is retained from the general method, and what does the biomedical data change?

- How can a patch representation contribute to a whole-slide prediction?
- Would pathology pretraining alone establish transfer to another laboratory?

Pathology pretraining

Histopathology patches; downstream regional and whole-slide tasks.

Source: [Chen et al., UNI study](#); see also the [authors' model repository](#).

What does a frozen-feature evaluation measure?

$$z = f_{\theta}(x), \quad p(y | x) = \text{softmax}(Wz + b)$$

Linear probe

Freeze θ ; fit W, b using task labels.

Fine-tuning

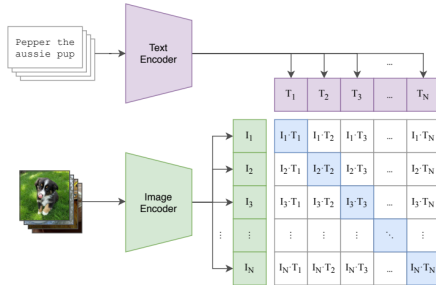
Update the encoder as well as the task mapping.

Discuss: Which comparison better isolates the usefulness of the original features?

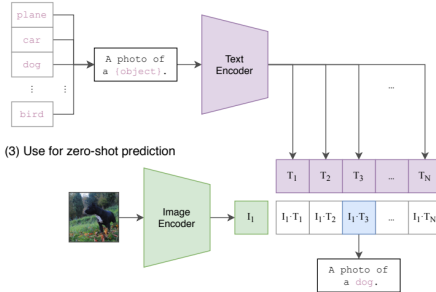
Source: [DINOv2, linear probing and fine-tuning evaluations](#).

How does language supervise vision?

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Discuss: What changes between training and zero-shot prediction?

Source: Radford et al., Figure 1; figure only.

What does the contrastive objective reward?

$$s_{ij} = \text{sim}(f(I_i), g(T_j)) , \quad \ell_i = -\log \frac{\exp(s_{ij}/\tau)}{\sum_j \exp(s_{ij}/\tau)}$$

I_i : image; T_j : candidate text; (I_i, T_i) : observed pair; $\tau > 0$: temperature.

- Which pair is rewarded, and against which alternatives?
- Can two different reports both be relevant to the same image?
- How does that complicate the interpretation of a mismatched pair in biomedicine?

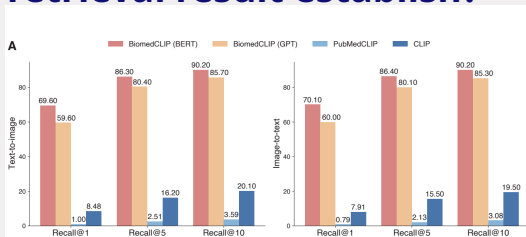
Source: CLIP, Section 2.2; image-to-text term shown.

CLIP and BiomedCLIP: what does adaptation change?

- Which biomedical image types and language patterns does the training collection cover?
- How would you distinguish a gain from domain coverage from a gain due to another model change?

Source: [Zhang et al., BiomedCLIP, dataset and domain adaptations.](#)

What does the retrieval result establish?

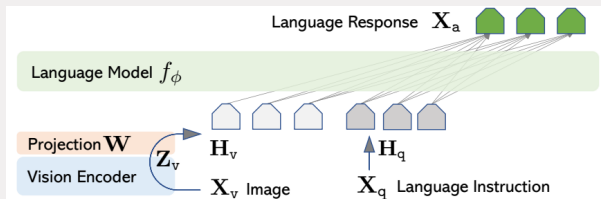


$$R@k = \frac{\text{queries with the paired item in the top } k}{\text{total queries}} \times 100\%$$

Discuss: Does finding the paired caption or figure establish clinical usefulness?

Source: [BiomedCLIP](#), Figure 2A; complete [retrieval-results](#) panel.

How does an image enter a language model?



Z_v : visual features; W : learned projection; $H_v = WZ_v$: visual vectors with the language model's embedding dimension.

Discuss: Trace these symbols in the figure. Does making visual information available guarantee that an answer uses it correctly?

Source: Liu et al., Figure 1; original LLaVA architecture.

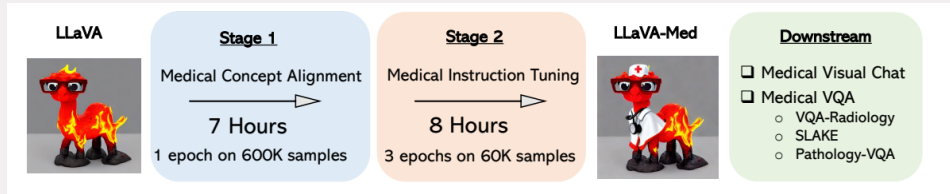
CLIP versus LLaVA: capabilities and use cases

	CLIP	LLaVA
Core capability	Match images and text	Generate text from image and prompt
Typical output	Similarity score or ranking	Caption, answer or dialogue
Typical uses	Retrieval; zero-shot classification	Visual questions; draft reports
Limitation	No native text generation	Generated claims need grounding checks

Discuss: When would you want an embedding model, and when would you want a generative assistant?

Source: [CLIP](#); [LLaVA](#).

LLaVA-Med: alignment and instruction following



- What does image-caption alignment teach that conversation examples may not?
- What behaviour does instruction tuning add?
- Why might either stage alone be insufficient for the intended task?

Source: Li et al., Figure 3; original curriculum diagram.

What does the response loss teach?

$$\mathcal{L} = - \sum_t \log p_{\theta}(y_t \mid I, q, y_{<t})$$

I : image; q : request; y : desired response.

- Where do the desired responses come from in the two generative papers?
- Which errors in those responses could become training supervision?
- Does improved response prediction demonstrate that the model uses the image?

Source: [LLaVA](#), generative training; compare [LLaVA-Med](#) instruction data.

Can a model answer well without looking?

Discuss the evaluation of generative visual assistants in the papers.

- What information does the model receive, and what information does the evaluator receive?
- How would you separate language fluency, factual support and visual grounding?
- What would comparing correct, missing and mismatched images reveal, and what would it not reveal?

Source: LLaVA-Med, Section 5.1; evaluator uses textual image context.

Compare the three model families

	Image-only	Contrastive image-text	Generative image-text
Main representation	Visual features	Aligned embeddings	Visual and language states
Training target	Derived from images	Matched pairs	Response tokens
Typical use	Task heads; similarity	Matching; retrieval; zero-shot classes	Captions; answers; reports

Discuss: Which distinctions are about the objective, and which are about downstream use? Where can the families overlap?

Choosing a role in a clinical workflow

- A pathologist wants to find visually related regions in a slide archive.
- A radiologist wants to retrieve images using a text description.
- A clinician wants an answer about an image or a draft report.

Discuss: Which representations and outputs fit each role? Where would human review enter? What would count as **useful performance**, beyond reproducing a benchmark score?